

ORB-30R Advanced

User, testing, and walk-forward manual

A practical operating guide for the free-plan-compatible TradingView research strategy, including setup, cost controls, weekly logging, and a disciplined walk-forward process.

READ THIS FIRST: The script is for research, backtesting, paper testing, and alert observation. It does not place broker orders. A TradingView strategy fill is a modeled fill, not an execution confirmation. One week of results cannot establish a durable edge.

Document control

Item	Value
Document date	12 September 2026
Source script	pinescript/RangeLab_ORB30R_Advanced.pine
Baseline chart	Five-minute standard candlesticks
Baseline state	Bar Magnifier disabled for free-plan compatibility
Instruments	ES and MES, analyzed independently

Immediate plan

- Archive this week's Strategy Tester report and List of Trades as Baseline v1 - Week 1.
- Do not change parameters because of this week's P&L.
- Run the same settings unchanged next week and record every session, including no-trade days.
- After enough history, use the rolling walk-forward design in Section 8.

The correct Pine source begins with `//@version=6`. Do not paste the MT5 file `RangeLab_HistoryExporter.mq5` into Pine Editor.

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Script: pinescript/RangeLab_ORB30R_Advanced.pine

Use: research, backtesting, paper testing, and alert observation only

Important boundary: This is not a broker execution system, does not place live orders, and does not prove profitability. A TradingView strategy fill is a modeled fill. Keep ES and MES results separate and treat one week of results as an observation, not as evidence of a durable edge.

1. What this script is

RangeLab ORB-30R Advanced is a TradingView Pine Script strategy implementing the frozen ORB-30R research specification. It tests a 30-minute opening-range breakout during the US cash session, with a trend filter, a range-width gate, stop-entry protection, ATR-based exits, integer contract sizing, modeled costs, and diagnostics.

The script is deliberately configured for the free TradingView plan. Bar Magnifier is disabled because TradingView charges for higher-detail historical intrabar calculation on plans that support it. The baseline therefore uses the chart's ordinary five-minute bars.

The strategy is intended to answer a narrow research question:

When the exact rules are applied consistently to an ES or MES five-minute series, what are the modeled entries, exits, costs, drawdowns, and out-of-sample results?

It is not intended to answer whether a broker will fill the order at the displayed price or whether future profits are guaranteed.

2. Frozen baseline rules

Use these rules for the baseline. Do not change them after seeing a result and still call that result an untouched out-of-sample result.

Rule	Baseline implementation
Chart	Five-minute, standard candlesticks
Clock	America/New_York
Cash session	09:30-16:00 ET, Monday-Friday
Opening range	09:30-10:00 ET; six completed five-minute bars
Entry window	10:00-13:30 ET
Exit	Submit close order during the 15:45 bar; intended fill at the 15:50 bar open
Trade count	Maximum one fill per session
Trend filter	Prior completed RTH close versus causal RTH EMA20 and EMA50
Volatility	Causal RTH Wilder ATR14

Warm-up	250 completed RTH bars
Range gate	Opening-range width must be no more than 5 ATR
Entry	One research tick beyond the opening-range high or low
Crossed-entry protection	Do not pretend to fill a stop order if the bar has already crossed it before the order is eligible
Initial stop	1.2 ATR from entry, rounded directionally to the tick grid
Target	2.0 ATR from entry, rounded directionally to the tick grid
Trailing	Disabled for the frozen baseline
Sizing	Integer contracts, equity-risk allowance, fee reserve, and slippage reserve
Bar Magnifier	Off for free-plan compatibility

The chart in the supplied screenshot was ES1!. That is a continuous futures symbol, not necessarily the exact contract that a broker would fill. Record the symbol and data source with every test. If possible, compare the continuous chart with the actual front-month contract used for paper or live observation.

3. Correct TradingView installation

1. Open TradingView in Chrome.
2. Open Pine Editor.
3. Create a new blank strategy.
4. Press Ctrl+A, then Delete. This avoids leaving the MT5 exporter in the editor.
5. Paste the complete contents of RangeLab_ORB30R_Advanced.pine.
6. The first line must be exactly:

```
//@version=6
```

7. The source must contain strategy(and use_bar_magnifier = false.
8. Click Save, then Add to chart.
9. Select a 5m chart and standard candles. Do not use Heikin Ashi, Renko, Range, Kagi, or another synthetic chart type.
10. Open Strategy Tester to Settings to Properties and verify initial capital, commission, slippage, and symbol settings.

Do not paste mt5/RangeLab_HistoryExporter.mq5. That is an MQL5 exporter for MetaTrader and will produce a translation error in Pine Editor. It contains lines such as #property strict and input ENUM_TIMEFRAMES; those lines do not belong in a Pine script.

What a healthy chart should show

- The strategy name appears in the chart legend.
- The status table appears at the upper right if Show status table is enabled.
- The table says 5m OK, not Use 5m.
- The opening-range high and low are plotted during RTH if enabled.
- Strategy Tester loads without a runtime error.
- There is no request to upgrade for Bar Magnifier.

If there are no trades, that does not automatically mean the script is broken. The 250-RTH-bar warm-up, range gate, trend gate, one-trade limit, and crossed-stop protection can legitimately reject a session.

4. Inputs and cost settings

Session and clock

Leave the timezone as America/New_York and use the default cash-session windows for the baseline. The chart's display timezone does not change the script's session calculations; the script uses the selected session timezone.

Risk and sizing

The default instrument is ES:

- ES point value: approximately \$50 per index point.
- MES point value: approximately \$5 per index point.
- Research tick size: 0.25 points.
- Default equity-risk allowance: 0.50 percent.
- Default maximum contracts: 1.

For an MES test, set:

- Sizing instrument = MES;
- Research tick size = 0.25;
- Round-trip fee reserve / contract to the fee assumption used in your test;
- Maximum integer contracts to the intended cap.

The fee reserve is used by the script when deciding whether an integer contract is affordable. TradingView's Strategy Properties commission is what affects modeled strategy P&L. Keep the fee-reserve input and the Properties-panel assumption consistent, and write both into your research log.

The slippage reserve is used for sizing. The strategy declaration also models one tick of slippage. These are separate safeguards, so document exactly what each one means in your report.

Baseline versus trailing experiment

Leave Enable separate trailing experiment off for the frozen baseline. If it is turned on, label that run as a different variant. Never mix its trades with the frozen baseline in one performance claim.

5. What to save from each test

A result is not reproducible unless its conditions are saved. For each ES or MES run, archive:

1. Strategy Tester Overview or Performance Summary.
2. Strategy Tester List of Trades export, if available.
3. A chart screenshot showing the symbol, timeframe, date range, and strategy name.
4. The complete input values.
5. Strategy Properties: initial capital, commission, slippage, and order settings.
6. Symbol details: ES or MES, continuous or dated contract, exchange/data source, and session setting.
7. The script version or a copy of the source file.
8. Any data gaps, TradingView warnings, runtime errors, or manual interventions.

Never overwrite an earlier report after changing an input. Give every run an ID, for example:

ES_5m_ORB30R_baseline_2026-09-12_v1

6. How to interpret this week's results

One week normally provides roughly five cash-session observations, and often fewer actual fills because the filters can skip sessions. That is far too small to estimate a stable win rate, expectancy, profit factor, or maximum drawdown.

Classify this week's result as:

Forward observation - sample too small for a performance conclusion.

A positive week is not a reason to increase size. A negative week is not a reason to change the rules. Both outcomes are useful if the execution and data were recorded correctly.

Before doing anything else, freeze and archive:

- the exact script source;
- the exact chart symbol and timeframe;
- the date range;
- all inputs;
- commission and slippage settings;
- the list of trades;
- the summary metrics;
- your notes about whether each order was actually observable or only modeled.

If the settings were fixed before the week began, the week can be labelled a genuine forward observation. If settings were changed after seeing earlier bars or trades, label it a post-hoc or exploratory result instead. Do not relabel it as clean out-of-sample evidence.

7. Walk-forward testing: the correct idea

Walk-forward testing separates information used to choose a rule from future information used to evaluate it.

- In-sample (IS): the historical window available when parameters are chosen or frozen.
- Out-of-sample (OOS): the next period, kept untouched until the choice is locked.
- Walk-forward: move the window forward, lock each choice, and evaluate the following unseen period.

A valid cycle is:

1. Choose a training window.
2. Use only that window to select or confirm parameters.
3. Lock the selected version and settings.
4. Run the next future window without changing anything.
5. Seal and archive the OOS result.
6. Move forward and repeat.
7. Combine only the sealed OOS results for the walk-forward record.

If you look at this week's P&L and then change the stop, target, range gate, or session before recording it, you have used the OOS result to train the next version. That is allowed as a research iteration, but it is no longer an untouched evaluation of the previous version.

8. Recommended plan from today

Because you already have this week's results, use the following sequence.

Stage 0 - Freeze this week now

Call the current script and settings Baseline v1. Do not optimize from this week's outcome. Archive the report and list of trades. Record whether this week was positive, negative, or flat only as a descriptive fact.

Stage 1 - Build a clean forward block

Run Baseline v1 unchanged for the next three to seven weeks. This gives a first forward block of approximately four to eight weeks including the week already observed. Record every session, including no-trade days.

At this stage the goal is operational validity:

- Did the script load without runtime errors?
- Was the chart really five-minute standard data?
- Were the sessions and exits at the intended New York times?
- Were there duplicate fills or missing fills?
- Were commission and slippage settings present?
- Was any manual intervention made?

Do not change parameters to improve the weekly total.

Stage 2 - First formal walk-forward cycle

Once enough history is available, use a pre-declared rolling design such as:

- 12 weeks IS, then 4 weeks OOS, roll forward by 4 weeks, or
- 8 weeks IS, then 2 weeks OOS when the available history is shorter.

The longer design is preferable for a one-trade-per-session system because it gives more trades in the training window. Neither design makes a small sample statistically certain; the purpose is to expose instability and look-ahead bias.

For each fold:

1. Freeze the current code version.
2. Use only the IS period for any permitted parameter selection.
3. Select a robust region, not the single best historical setting.
4. Lock the chosen settings before the OOS start date.
5. Run OOS with no edits.
6. Archive the OOS report separately.
7. Roll forward and repeat.

The frozen ORB-30R baseline should remain as a permanent benchmark in every fold. If you test variants, compare each variant against that benchmark rather than replacing it.

Stage 3 - Promotion decision

Do not promote from paper observation to larger size because of one good week. Require:

- valid and reproducible data;
- no unresolved script or timing errors;
- enough completed sessions and trades for the chosen metrics to be defined;

- results after modeled commission and slippage;
- drawdown within a pre-declared risk budget;
- performance that is not dependent on one unusually large trade;
- similar behavior across more than one OOS fold;
- ES and MES reviewed independently;
- no evidence that the parameter choice used future OOS information.

If a metric is undefined because there are too few trades, report it as undefined. Do not replace it with a zero or a favorable estimate.

9. What may and may not be changed

Changes that create a new version

Any change to these creates a new version and requires a new research log entry:

- session times or timezone;
- opening-range length;
- entry window;
- EMA or ATR lengths;
- warm-up length;
- range-gate multiple;
- entry tick distance;
- stop or target multiple;
- trailing logic;
- sizing formula or maximum contracts;
- commission, slippage, or fee assumptions;
- chart type, timeframe, or symbol;
- code logic.

Safe operational corrections

Fixing a copy/paste error, removing the old MQL5 script from Pine Editor, or setting Bar Magnifier to false to make the free-plan script run is an operational correction. Still document it and restart the affected test if the error could have changed trades.

Do not optimize all inputs at once

If a formal research variant is allowed, pre-declare a small search space and an objective measured after costs. Prefer a stable plateau of nearby settings to a single best point. Keep the frozen baseline unchanged so the result can be compared with the original specification.

10. Daily and weekly journal template

Record one row per session, including sessions with no trade.

Field	What to record
Date	New York cash-session date
Instrument	ES or MES, kept separate

Symbol	For example ES1! or the dated contract
Data source	TradingView exchange/vendor feed
OR high / low	Frozen 09:30-10:00 range
OR width	Price points and width divided by prior ATR
Prior ATR	Causal RTH ATR used by the gate
Trend	Long, short, or rejected
Trigger	Rounded stop-entry price
Quantity	Integer contracts or zero
Order result	Armed, filled, cancelled, or rejected
Entry	Modeled fill price and time
Stop / target	Frozen bracket values
Exit	Price, time, and exit reason
Costs	Commission and slippage assumption
Result	Dollars and R multiple
Notes	Gaps, warnings, manual action, or data issue

At the end of each week, save the raw journal and a read-only summary. Do not edit the historical row after a later result is known; append a correction note instead.

11. Metrics to review after each sealed block

Use the same definitions every time:

- number of eligible sessions;
- number of fills;
- win rate;
- average win and average loss;
- expectancy per trade in dollars and R;
- profit factor, when there are both gains and losses;
- net P&L after modeled costs;
- maximum peak-to-trough drawdown;
- worst day and worst week;
- average holding time;
- percentage of sessions with no trade;
- long and short results separately;
- ES and MES results separately;
- IS versus OOS comparison;
- results by walk-forward fold.

The most important first check is not the headline profit. It is whether the OOS trades were generated by the same rules, symbol conditions, session settings, and cost assumptions as the frozen baseline.

12. Decision matrix for the current week

Observation	Action
Positive or negative week, but fewer than about 20-30 valid fills	Mark insufficient sample; continue unchanged
Runtime error, wrong timeframe, wrong source file, or wrong session	Correct the setup; restart the affected measurement block
Good result but settings were changed after observing the week	Label exploratory; do not count as clean OOS
Negative result with valid unchanged settings	Keep it in the OOS record; do not optimize reactively
Results vary sharply between ES and MES	Keep them separate; investigate contract, costs, and liquidity differences
Results depend on one unusually large trade	Continue testing; do not promote on that week alone
Repeated OOS deterioration across several folds	Pause promotion and diagnose; do not simply widen the search until something looks good

These are research controls, not a promise that any outcome will be profitable.

13. What to send for the next review

To review this week's result without contaminating the next test, provide:

1. The Strategy Tester summary screenshot or export.
2. The List of Trades export or a table of the trades.
3. The symbol, timeframe, date range, and data source.
4. All script input values.
5. Commission and slippage Properties settings.
6. Whether the script was unchanged before and during the week.
7. Any no-trade sessions or runtime warnings.

The result can then be classified as valid forward data, exploratory data, or invalid setup data. It should not be merged with another instrument or another script version.

Immediate next actions

1. Save this week's report and list of trades as Baseline v1 - Week 1.
2. Do not change the ORB-30R inputs because of this week's P&L.
3. Run the same ES or MES setup unchanged next week on the five-minute chart.
4. Add one journal row per cash session, including no-trade sessions.
5. After the first four-week block, perform a data and execution audit only.
6. After enough history is collected, begin the pre-declared 12-week IS / 4-week OOS rolling walk-forward.
7. Keep the frozen baseline and every variant in separate folders and separate reports.

Bottom line: This week's result is valuable as the first forward observation, but it is not yet a reason to change the model or increase risk. Preserve it, continue unchanged, and let future sealed blocks determine whether the behavior is repeatable.